ReWaste

An AI-Powered Industrial Waste Exchange Platform

SDG 9: Industry, Innovation & Infrastructure | **SDG 12: Responsible Consumption & Production**

A B S T R A C T

ReWaste is a smart industrial waste exchange platform designed to connect industries that generate reusable waste with industries that can utilize the same waste as raw material. The system promotes sustainable industrial practices by reducing landfill disposal and encouraging resource reuse through a digital waste-sharing network.

In many industries, large amounts of reusable waste materials such as plastics, metals, chemicals, paper scraps, wood dust, textile residue, and organic by-products are discarded due to the lack of proper coordination between industries. At the same time, other manufacturing units spend significant amounts of money purchasing raw materials that could potentially be replaced by these reusable industrial by-products. ReWaste addresses this gap by creating a centralized platform where industries can digitally exchange reusable waste resources efficiently and responsibly.

Industries can register on the platform, create company profiles, upload waste details such as material type, quantity, location, availability, and images, and search for reusable industrial by-products from other companies. The platform includes a rule-based matching engine that analyzes waste categories and industrial requirements to recommend suitable exchanges between companies without relying on expensive third-party AI services or premium APIs.

The system also provides advanced filtering and search functionalities that allow industries to identify suitable waste materials based on category, location, quantity, and industrial usage. Companies can send exchange requests, manage approvals, track transaction status, and maintain records of completed exchanges through an integrated waste management dashboard.

ReWaste includes an administrative monitoring module that helps verify industries, manage waste listings, monitor platform activities, and generate sustainability reports. The platform also features sustainability analytics that display estimated landfill reduction, waste reused, carbon emission reduction, and resource savings achieved through industrial collaboration.

To ensure affordability and accessibility, the project is designed using low-cost and open-source technologies without depending on premium APIs or paid mapping services. Nearby industry matching can be achieved through location-based filtering using district and state information along with basic distance calculation techniques.

By enabling industries to treat waste as a reusable economic resource rather than disposable material, ReWaste supports the concept of a circular economy and industrial symbiosis. The

platform helps industries reduce disposal costs, optimize resource utilization, improve environmental sustainability, and encourage responsible industrial production practices.

ReWaste aligns with Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by promoting innovative industrial collaboration and digital infrastructure, and Sustainable Development Goal 12 (Responsible Consumption and Production) by encouraging waste reduction, recycling, and sustainable resource management practices.

PROPOSED MODULES

1. User Registration and Authentication Module

Allows industries to create accounts, log in securely, and manage company profiles.

2. Waste Upload and Management Module

Enables companies to upload waste details such as material type, quantity, location, availability, and images.

3. Waste Search and Filtering Module

Allows industries to search and filter available industrial waste based on category, quantity, and location.

4. Smart Matching Module

Implements a rule-based matching system that recommends suitable industries capable of reusing specific waste materials.

5. Exchange Request and Approval Module

Allows industries to send, receive, accept, or reject waste exchange requests and manage transaction status.

6. Sustainability Analytics Dashboard

Displays waste reuse statistics, estimated landfill reduction, carbon reduction estimates, and exchange reports.

7. Admin Monitoring Module

Allows administrators to verify industries, monitor platform activities, manage waste listings, and generate reports.

KEYWORDS

Industrial waste exchange · Circular economy · Waste matchmaking · Sustainability analytics · SDG 9 · SDG 12 · Rule-based engine · Waste-to-resource

PROJECT DETAILS

Project Title: WasteMatch — An AI-Powered Industrial Waste Exchange Platform

Project Type: Full-Stack Web Application

SDG Alignment: SDG 9 (Industry, Innovation & Infrastructure) | SDG 12 (Responsible Consumption & Production)

Tech Stack: React.js · Node.js / Express · PostgreSQL · Python ·

Matching Engine: Rule-based waste category and industry compatibility filtering

Target Users: Industrial companies, SMEs, waste producers, raw material consumers

Key Deliverable: Live platform with impact dashboard, matching engine, and company portal